Example 10: (A Constructive Existence Proof) Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

Solution: After considerable computation (such as a computer search) we find that $1729 = 10^3 + 9^3 = 12^3 + 1^3$.

- Because we have displayed a positive integer that can be written as the sum of cubes in two different ways, we are done. □

Example 11: (A Nonconstructive Existence Proof) Show that there exists irrational numbers x and y such that x^y is rational.

Solution: $P(x)$: x is irrational $Q(y)$: y is irrational $R(x,y)$: x^y is rational.

We need to prove that $\boxed{\exists x \exists y (P(x) \land Q(y) \land R(x,y)) \text{ is True.}}$

Suppose, for the sake of contradiction, $\neg \exists x \exists y (P(x) \land Q(y) \land R(x,y))$ is True.
By De Morgan's Law of Quantifiers, $\forall x \neg \exists y (P(x) \land Q(y) \land R(x,y))$ is True.
By De Morgan's Law of Quantifiers, $\forall x \forall y \neg (P(x) \land Q(y) \land R(x,y))$ is True.
By De Morgan's Law, $\forall x \forall y (\neg P(x) \lor \neg Q(y) \lor \neg R(x,y))$ is True.
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Solution from the Book (Brilliant)

- By Example 11 in Section 1.7 we know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. If it is rational, we have two irrational numbers x and y with x^y rational, namely, $x = \sqrt{2}$ and $y = \sqrt{2}$. On the other hand if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^{\sqrt{2} \cdot \sqrt{2}} = (\sqrt{2})^2 = 2$
- This proof is an example of a nonconstructive existence proof because we have not found irrational numbers x and y such that x^y is rational. Rather we have shown that either the pair $x = \sqrt{2}$, $y = \sqrt{2}$ or the pair $x = \sqrt{2}^{\sqrt{2}}$, $y = \sqrt{2}$ have the desired property, but we do not know